Supplementary Material for:

Martingale Innovations from Contractive Recurrent Networks and Dimension-Robust Change-Point Detection

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A. Background and Preliminaries

We work throughout on a complete probability space $(\Omega, \mathcal{A}, \mathbb{P})$. All convergence statements are in the almost-sure or L^p sense unless otherwise noted.

A.1. Filtrations and the Information Flow

Definition A.1 (Filtration and natural filtration). A *filtration* is an increasing family of sub-σ-algebras $\mathcal{F}_1 \subseteq \mathcal{F}_2 \subseteq \cdots \subseteq \mathcal{A}$. For a sequence $(X_t)_{t \geq 1}$ of random variables, the *natural filtration* is $\mathcal{F}_t = \sigma(X_1, \ldots, X_t)$ and its completion $\mathcal{F}_\infty^+ = \sigma\big(\bigcup_{t \geq 1} \mathcal{F}_t\big)$.

A *decreasing filtration* (or *reverse filtration*) is a family $\mathcal{G}_1 \supseteq \mathcal{G}_2 \supseteq \cdots$ of sub-σ-algebras. The canonical example relevant to this paper is

$$\mathcal{G}_n = \sigma(X_n, X_{n+1}, \ldots), \tag{A.1}$$

which formalizes the information contained in the *tail* of the sequence starting at index n.

Definition A.2 (Tail σ-algebra). The *tail σ-algebra* of a sequence $(X_t)_{t \geq 1}$ is

$$\mathcal{T}_\infty = \bigcap_{n=1}^{\infty} \sigma(X_n, X_{n+1}, \ldots) = \bigcap_{n=1}^{\infty} \mathcal{G}_n.$$

An event $A \in \mathcal{T}_\infty$ is *tail-measurable*: its occurrence or non-occurrence is not affected by any finite initial segment of the sequence.

Example A.3 (Tail-measurable functionals). For an i.i.d. or ergodic sequence (X_t): (a) the limit $\lim_{n\to\infty} n^{-1} \sum_{t=1}^{n} X_t$ (when it exists) is $\mathcal{T}_\infty$-measurable; (b) the event $\{\limsup_{t\to\infty} X_t > c\}$ is tail-measurable; (c) the population mean $\mu = \mathbb{E}[X_1]$ is a constant, hence trivially $\mathcal{T}_\infty$-measurable under any probability measure. Finite-dimensional functions of $(X_1, \ldots, X_k)$ for fixed k are *not* tail-measurable in general.

A.2. Martingales and Reverse Martingales

Definition A.4 (Martingale and martingale difference sequence). A sequence $(M_t, \mathcal{F}_t)_{t\geq 1}$ is a *martingale* if $\mathbb{E}[|M_t|] < \infty$ and $\mathbb{E}[M_{t+1} \mid \mathcal{F}_t] = M_t$ a.s. for all t. The sequence $(D_t)_{t\geq 1}$ is a *martingale difference sequence* (MDS) if $\mathbb{E}[|D_t|] < \infty$ and $\mathbb{E}[D_t \mid \mathcal{F}_{t-1}] = 0$ a.s. for all $t \geq 2$.

The posterior mean $\mathbb{E}[\theta \mid \mathcal{F}_t]$, for a fixed integrable parameter θ and observations $X_1, X_2, \ldots$, is the canonical martingale in Bayesian sequential analysis.

Definition A.5 (Reverse martingale). Let $(\mathcal{G}_n)_{n\geq 1}$ be a decreasing filtration. A sequence $(Y_n, \mathcal{G}_n)_{n\geq 1}$ is a *reverse martingale* if $\mathbb{E}[|Y_n|] < \infty$ and

$$\mathbb{E}[Y_n \mid \mathcal{G}_{n+1}] = Y_{n+1} \quad \text{a.s. for all } n \geq 1.$$

A prototypical reverse martingale is obtained by taking an integrable random variable Y and setting $Y_n = \mathbb{E}[Y \mid \mathcal{G}_n]$ for a decreasing filtration $(\mathcal{G}_n)$. Exchangeability arguments also represent empirical averages through appropriate decreasing sigma-fields. The key convergence theorem is:

Theorem A.6 (Reverse martingale convergence; Doob 1953). *If $(Y_n, \mathcal{G}_n)_{n\geq 1}$ is an L^1-bounded reverse martingale with decreasing filtration $(\mathcal{G}_n)$, then*

$$Y_n \xrightarrow{a.s.\text{ and } L^1} Y_\infty := \mathbb{E}[Y_1 \mid \mathcal{G}_\infty], \qquad \mathcal{G}_\infty = \bigcap_{n\geq 1} \mathcal{G}_n.$$

Proof. Let $(Y_n, \mathcal{G}_n)_{n\geq 1}$ be an L^1-bounded reverse martingale with decreasing filtration $\mathcal{G}_1 \supseteq \mathcal{G}_2 \supseteq \cdots$ and $\mathcal{G}_\infty = \bigcap_{n\geq 1} \mathcal{G}_n$. Without loss of generality, write $Y_n = \mathbb{E}[Y_1 \mid \mathcal{G}_n]$, which is consistent with the reverse martingale property by the tower law: for $m > n$, $\mathcal{G}_m \subseteq \mathcal{G}_n$ implies $\mathbb{E}[Y_n \mid \mathcal{G}_m] = \mathbb{E}[\mathbb{E}[Y_1 \mid \mathcal{G}_n] \mid \mathcal{G}_m] = \mathbb{E}[Y_1 \mid \mathcal{G}_m] = Y_m$ a.s.

Almost-sure convergence. For any $a < b$, let $U_n([a,b])$ be the number of upcrossings of $[a,b]$ by the sequence $Y_1, Y_2, \ldots, Y_n$. Doob's upcrossing inequality for reverse martingales (Neveu, 1975, Chapter V) gives

$$(b-a)\,\mathbb{E}[U_n([a,b])] \leq \mathbb{E}[(Y_1 - a)^+] \leq \mathbb{E}[|Y_1|] + |a| < \infty.$$

Letting $n \to \infty$, the monotone convergence theorem gives $\mathbb{E}[U_\infty([a,b])] < \infty$, so $U_\infty([a,b]) < \infty$ a.s. Since $a < b$ are arbitrary rationals, the number of upcrossings of every rational interval is finite almost surely, which implies $Y_n \to Y_\infty$ a.s. for some $[-\infty, \infty]$-valued random variable Y_∞.

L^1 convergence and integrability of the limit. For each n, $|Y_n| = |\mathbb{E}[Y_1 \mid \mathcal{G}_n]| \leq \mathbb{E}[|Y_1| \mid \mathcal{G}_n]$ by Jensen's inequality. The family $\{\mathbb{E}[|Y_1| \mid \mathcal{G}_n]\}_{n\geq 1}$ is uniformly integrable by the standard fact that the collection of conditional expectations $\{\mathbb{E}[Z \mid \mathcal{G}] : \mathcal{G} \subseteq \mathcal{A}\}$ of a fixed $Z \in L^1$ is uniformly integrable (Williams, 1991, Sec. 13.4). Hence $\{Y_n\}$ is uniformly integrable. Uniform integrability together with a.s. convergence implies L^1 convergence: $\|Y_n - Y_\infty\|_{L^1} \to 0$, and in particular $Y_\infty \in L^1$.

Identification of the limit. For any $A \in \mathcal{G}_\infty \subseteq \mathcal{G}_n$, the definition of conditional expectation gives $\int_A Y_n \, d\mathbb{P} = \int_A Y_1 \, d\mathbb{P}$ for all n. Passing to the limit using L^1 convergence: $\int_A Y_\infty \, d\mathbb{P} = \int_A Y_1 \, d\mathbb{P}$ for all $A \in \mathcal{G}_\infty$. By the definition of conditional expectation, $Y_\infty = \mathbb{E}[Y_1 \mid \mathcal{G}_\infty]$ a.s. □

Remark A.7 (Comparison with forward convergence). Doob's *forward* martingale convergence theorem requires L^1-boundedness and yields almost-sure convergence but not necessarily L^1 convergence (unless the martingale is uniformly integrable). Reverse martingales are automatically uniformly integrable, so both modes of convergence are free. This property is useful for tail-projection arguments such as Lemma 2.1 of the main paper.

A.3. The Hewitt–Savage Zero-One Law

Theorem A.8 (Hewitt–Savage; Hewitt and Savage 1955). *Let $(X_t)_{t\geq 1}$ be an i.i.d. sequence. Then every event invariant under finite permutations of the coordinates has probability 0 or 1. In particular, every tail event $A \in \mathcal{T}_\infty$ satisfies $\mathbb{P}(A) \in \{0,1\}$.*

For i.i.d. sequences, Theorem A.8 implies that the tail σ-algebra is trivial, so $\mathbb{E}[Z \mid \mathcal{T}_\infty] = \mathbb{E}[Z]$ for any integrable Z. The statement should not be extended to arbitrary exchangeable sequences: by de Finetti's theorem, an exchangeable but non-i.i.d. Bernoulli sequence can have a nontrivial tail generated by the random mixing parameter. For stationary ergodic sequences, the ergodic theorem identifies limits of time averages, but it does not imply that every tail-measurable functional is literally a time average.

A.4. Recurrent Neural Networks and LSTMs

We give precise definitions of the two architectures whose hidden states we analyze.

Definition A.9 (Recurrent Neural Network). An *RNN* processes a sequence $(x_t)_{t=1}^T$, $x_t \in \mathbb{R}^p$, via

$$h_t = \phi(W_h h_{t-1} + W_x x_t + b_h), \tag{A.2}$$

where $h_t \in \mathbb{R}^d$ is the *hidden state*, $W_h \in \mathbb{R}^{d\times d}$, $W_x \in \mathbb{R}^{d\times p}$, $b_h \in \mathbb{R}^d$, and $\phi : \mathbb{R}^d \to \mathbb{R}^d$ is a coordinate-wise non-linear activation (e.g. tanh or ReLU). The initial state $h_0 \in \mathbb{R}^d$ is fixed or learnable; the L^2 theory below assumes it is square-integrable.

Remark A.10 (Notation). In this appendix d denotes the hidden-state dimension and p the input dimension. In the main paper (Section 2), the same dimensions are written p (hidden) and d_{in} (input); the symbol d in the main paper refers to the observation dimension in the empirical studies (Section 4).

Assumption A.11 (Effective operator-norm contractivity). Throughout the paper we assume

$$L_\phi \, \|W_h\|_{\text{op}} < 1,$$

where L_ϕ is the Lipschitz constant of the activation ϕ and $\|W_h\|_{\text{op}} = \sigma_{\max}(W_h)$ is the operator norm. This is the contraction needed for L^2-stability of (A.2). The spectral radius alone is insufficient for non-normal matrices because it does not control the vector norm. Spectral normalization can encourage or enforce this condition after accounting for L_ϕ (Miyato et al., 2018).

Definition A.12 (Long Short-Term Memory). An *LSTM* (Hochreiter and Schmidhuber, 1997) augments the basic RNN with a cell state $c_t \in \mathbb{R}^d$ and three gating mechanisms:

$$f_t = \sigma(W_f[h_{t-1}; x_t] + b_f), \tag{A.3}$$
$$i_t = \sigma(W_i[h_{t-1}; x_t] + b_i), \tag{A.4}$$
$$o_t = \sigma(W_o[h_{t-1}; x_t] + b_o), \tag{A.5}$$
$$\tilde{c}_t = \tanh(W_c[h_{t-1}; x_t] + b_c), \tag{A.6}$$
$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t, \tag{A.7}$$
$$h_t = o_t \odot \tanh(c_t), \tag{A.8}$$

where $[h_{t-1}; x_t] \in \mathbb{R}^{d+p}$ denotes concatenation, σ is the logistic sigmoid, and $\odot$ is coordinate-wise multiplication. The *forget gate* $f_t \in (0,1)^d$ controls how much of the previous cell state survives to time t.

Remark A.13 (Contractivity of LSTM). The coordinate-wise bound $f_t \in (0,1)^d$ does not by itself make the full LSTM cell recursion uniformly contractive. The gate can be arbitrarily close to one, and f_t depends on h_{t-1}, which depends on c_{t-1}, so derivatives of the gate map also enter the Lipschitz constant. A rigorous contraction theorem for LSTMs therefore requires a separate assumption, such as a uniform gate bound $\|f_t\|_\infty \le \gamma < 1$ together with Lipschitz control of the gate maps. In this paper, the rigorous contraction theorem is stated only for the basic RNN recurrence.

A.5. Stationary Ergodic Processes

Definition A.14 (Stationarity and ergodicity). A process $(X_t)_{t\ge1}$ is *(strictly) stationary* if

$$(X_{t+k})_{t\ge1} \stackrel{d}{=} (X_t)_{t\ge1}$$

for all $k \ge 1$. It is *ergodic* if every shift-invariant event has probability 0 or 1. By the ergodic theorem (Walters, 1982), for any L^1 functional g,

$$\frac{1}{T}\sum_{t=1}^{T} g(X_t) \xrightarrow{a.s.} \mathbb{E}[g(X_1)] \quad \text{as } T \to \infty.$$

Stationarity is the maintained assumption throughout the theory sections (Sections 2–3) of the main paper; the empirical Section 4 deliberately includes non-stationary change-point and real-data settings. The non-stationary extension via locally stationary processes (Dahlhaus, 1997) is treated in Section 2.3 of the main paper.

B. Proofs

For reference, Table B.1 maps each result of the main paper to the object it concerns, the assumptions it requires, and what it establishes; the proofs follow in order.

B.1. Proof of Lemma 2.1 of the main paper (Equivalence of Forward and Tail Limits)

Proof of Lemma 2.1 in the main paper. **Part (a).** Since $\theta \in L^\infty \subseteq L^1$ and θ is $\mathcal{T}_\infty$-measurable, it is also $\mathcal{F}_\infty^+$-measurable because $\mathcal{T}_\infty \subseteq \mathcal{G}_1 = \sigma(X_1, X_2, \ldots) \subseteq \mathcal{F}_\infty^+$. By Doob's L^1 martingale convergence theorem (Doob, 1953), $\mathbb{E}[\theta \mid \mathcal{F}_t] \to \mathbb{E}[\theta \mid \mathcal{F}_\infty^+]$ a.s. and in L^1. Since θ is $\mathcal{F}_\infty^+$-measurable, $\mathbb{E}[\theta \mid \mathcal{F}_\infty^+] = \theta$ a.s.

Part (b). The sequence $(\mathbb{E}[\theta \mid \mathcal{G}_n], \mathcal{G}_n)_{n\ge1}$ is a reverse martingale: for $m > n$, $\mathcal{G}_m \subseteq \mathcal{G}_n$ implies $\mathbb{E}[\mathbb{E}[\theta \mid \mathcal{G}_n] \mid \mathcal{G}_m] = \mathbb{E}[\theta \mid \mathcal{G}_m]$ by the tower property. Boundedness of θ gives L^1-boundedness. By the reverse martingale convergence theorem (Theorem A.6; Doob 1953), $\mathbb{E}[\theta \mid \mathcal{G}_n] \to \mathbb{E}[\theta \mid \mathcal{T}_\infty]$ a.s. and in L^1. Since θ is $\mathcal{T}_\infty$-measurable, $\mathbb{E}[\theta \mid \mathcal{T}_\infty] = \theta$ a.s.

Part (c). Immediate from (a) and (b): both limits equal θ a.s.

Table B.1. Map of the main-paper results. The top block (tier one) concerns the contractive RNN and its innovations; the bottom block (tier two) concerns the Innovation CUSUM detector and requires no RNN contraction theorem, only conditional residual-drift and tail conditions on fitted residuals.

Result	Object	Key assumptions	Establishes
Thm 2.5	Contractive RNN state h_t	stationary ergodic input, $\mathbb{E}\,\|X\|^2 < \infty$, $L_\phi\,\|W_h\|_{\text{op}} < 1$	Doob/MDS decomposition; geometric innovation stabilization
Cor 2.8	RNN innovations M_t	additionally $(2+\delta)$-moment, $\Sigma_M^* \succ 0$	functional CLT for cumulative innovations
Prop 2.9	Prediction residuals e_t	Bayes predictor (exact); L^2-accurate fitted predictor (approx.)	exact / approximate MDS
Prop 2.10	Recurrent predictor residuals	hypotheses of Thm 2.5; Lipschitz readout	residual drift $\le L_o\rho^t\Delta_0$ + stationary readout error
Prop 2.11	Fitted vs. linear residuals	nonlinear conditional mean; consistent learned predictor	irreducible linear defect; learned defect $\to 0$
Thm 3.2	Innovation CUSUM score	drift margin $k > \eta$, conditional sub-Gaussianity	exponential in-control ARL lower bound
Thm 3.6	Innovation CUSUM score	post-change drift $\ge k+\mu$, bounded increments	out-of-control detection-delay upper bound
Cor 3.7	Innovation CUSUM score	Assumptions above	exponential-ARL / logarithmic-delay trade-off
Cor 3.9	Mahalanobis innovation score	uniform-in-d: bounded η_d, $\lambda_{\min}(\Sigma_0) \ge c_0$, conditional homoskedasticity, $O(1)$ score sub-Gaussianity	dimension-stable ARL

Parts (d) and (e). For general $\theta \in L^1$, part (a) extends to $\mathbb{E}[\theta \mid \mathcal{F}_t] \to \mathbb{E}[\theta \mid \mathcal{F}_\infty^+]$ by the same Doob argument. For part (e), the reverse martingale convergence in (b) applies unchanged since the reverse martingale is $(\mathbb{E}[\theta \mid \mathcal{G}_n], \mathcal{G}_n)$, and the limit is $\mathbb{E}[\theta \mid \mathcal{T}_\infty]$ regardless of whether θ is tail-measurable. The two limits agree if and only if $\mathbb{E}[\theta \mid \mathcal{F}_\infty^+]$ is $\mathcal{T}_\infty$-measurable, which holds when θ itself is tail-measurable. □

B.2. Proof of Theorem 2.5 of the main paper (MDS Decomposition with Stabilization Rate)

Proof of Theorem 2.5 in the main paper. Throughout, $\rho = L_\phi\,\|W_h\|_{\text{op}} < 1$ and $\Delta_0 = \left\|h_0 - H_0^*\right\|_{L^2}$.

Part (i): L^2-boundedness. We prove the C_0 formula for the case $\phi(0) = 0$ (e.g. tanh, ReLU), where the Lipschitz bound gives $\|\phi(u)\| \le L_\phi\,\|u\|$. (For activations with $\phi(0) \neq 0$, e.g. sigmoid, an additional term $L_\phi\,\|\phi(0)\|\,/(1-\rho)$ is added to C_0, consistent with the general formula stated in Theorem 2.5 of the main paper.) By the Lipschitz bound $\|\phi(u)\| \le L_\phi\,\|u\|$ and Minkowski's inequality in L^2,

$$\|h_t\|_{L^2} \le L_\phi\,\|W_h h_{t-1} + W_x X_t + b_h\|_{L^2} \le \rho\,\|h_{t-1}\|_{L^2} + L_\phi\,\|W_x\|_{\text{op}}\,\|X_t\|_{L^2} + L_\phi\,\|b_h\|\,.$$

Let $c := L_\phi(\|W_x\|_{\text{op}}\,\|X_1\|_{L^2} + \|b_h\|)$, which is finite by stationarity. Iterating the inequality and summing the geometric series gives

$$\|h_t\|_{L^2} \le \rho^t\,\|h_0\|_{L^2} + \frac{c}{1-\rho},$$

so $\sup_t \|h_t\|_{L^2} \le \|h_0\|_{L^2} + c/(1-\rho) =: C_0 < \infty$.

Part (ii): MDS structure. Since h_t is $\mathcal{F}_t$-measurable and square-integrable by part (i), the conditional expectation $D_t := \mathbb{E}[h_t \mid \mathcal{F}_{t-1}]$ is well-defined and $\mathcal{F}_{t-1}$-measurable. Setting $M_t := h_t - D_t$,

$$\mathbb{E}[M_t \mid \mathcal{F}_{t-1}] = \mathbb{E}[h_t - D_t \mid \mathcal{F}_{t-1}] = D_t - D_t = 0 \quad \text{a.s.}$$

Part (iii): Innovation L^2 bound. Conditional on $\mathcal{F}_{t-1}$, the previous state h_{t-1} is fixed and the only new random input is X_t. Put $\mu_t = \mathbb{E}[X_t \mid \mathcal{F}_{t-1}]$ and define $g_t(x) := \phi(W_h h_{t-1} + W_x x + b_h)$. Since h_{t-1} is $\mathcal{F}_{t-1}$-measurable, $g_t(\mu_t) = \phi(W_h h_{t-1} + W_x \mu_t + b_h)$ is $\mathcal{F}_{t-1}$-measurable. By the best-approximation property of the conditional expectation ($D_t = \mathbb{E}[h_t|\mathcal{F}_{t-1}]$ minimizes the L^2 distance to h_t over all $\mathcal{F}_{t-1}$-measurable functions),

$$\begin{aligned}\mathbb{E}\left\{\|M_t\|^2 \mid \mathcal{F}_{t-1}\right\} &\le \mathbb{E}\left\{\|g_t(X_t) - g_t(\mu_t)\|^2 \mid \mathcal{F}_{t-1}\right\} \\ &\le L_\phi^2 \, \|W_x\|_{\mathrm{op}}^2 \, \mathbb{E}\left\{\|X_t - \mu_t\|^2 \mid \mathcal{F}_{t-1}\right\}.\end{aligned}$$

Taking unconditional expectations: since $\mathbb{E}\{\|X_t - \mu_t\|^2 \mid \mathcal{F}_{t-1}\} = \mathrm{Var}(X_t \mid \mathcal{F}_{t-1})$ is the conditional variance, the law of total variance gives $\mathbb{E}[\mathrm{Var}(X_t \mid \mathcal{F}_{t-1})] = \mathrm{Var}_2(X_t) - \mathbb{E}\,\|\mu_t - \mathbb{E}X_t\|^2 \le \mathrm{Var}_2(X_t)$, which yields equation (1) in the main paper.

Part (iv): Innovation stabilization rate. Write $M_t - M_t^* = (h_t - H_t^*) - (D_t - D_t^*)$. By the triangle inequality, $\left\|M_t - M_t^*\right\|_{L^2} \le \left\|h_t - H_t^*\right\|_{L^2} + \left\|D_t - D_t^*\right\|_{L^2}$. The coupling bound (Proposition B.1 below) gives $\left\|h_t - H_t^*\right\|_{L^2} \le \rho^t \Delta_0$. Since conditional expectation is an L^2 contraction, $\left\|D_t - D_t^*\right\|_{L^2} = \left\|\mathbb{E}[h_t - H_t^* \mid \mathcal{F}_{t-1}]\right\|_{L^2} \le \left\|h_t - H_t^*\right\|_{L^2} \le \rho^t \Delta_0$. Adding the two bounds gives $\left\|M_t - M_t^*\right\|_{L^2} \le 2\rho^t \Delta_0$.

Part (v): Variance decomposition and Σ_M^*. The law of total variance applied coordinatewise, combined with $\mathbb{E}[M_t \mid \mathcal{F}_{t-1}] = 0$, gives

$$\mathbb{E}\,\|h_t - \mathbb{E}h_t\|^2 = \mathbb{E}\,\|D_t - \mathbb{E}h_t\|^2 + \mathbb{E}\,\|M_t\|^2,$$

since the cross term $2\mathbb{E}\langle D_t - \mathbb{E}h_t, M_t\rangle = 0$.

For the stationary regime, recall that the filtration is the two-sided past $\mathcal{F}_{t-1} = \sigma(X_s : s \le t-1) \vee \sigma(h_0)$ with h_0 independent of the process (main paper, Section 2 setup). By the causal construction (Proposition B.1), H_t^* is the same fixed measurable functional of $(\ldots, X_{t-1}, X_t)$ at every t, and since $\sigma(h_0)$ is independent of $\sigma(X_s : s \in \mathbb{Z})$, conditioning on it leaves the conditional expectation unchanged: $D_t^* = \mathbb{E}[H_t^* \mid \mathcal{F}_{t-1}] = \mathbb{E}[H_t^* \mid \sigma(X_s : s \le t-1)]$ is likewise the same functional of the shifted path at every t. Hence $(M_t^* = H_t^* - D_t^*)_{t\in\mathbb{Z}}$ is strictly stationary and $\Sigma_M^* := \mathbb{E}[M_t^*(M_t^*)^\top]$ is independent of t. (With the one-sided filtration $\sigma(X_1, \ldots, X_{t-1})$ this would fail: the conditioning set is not shift-invariant.) Positive semi-definiteness is immediate. The bound $\Sigma_M^* \preceq \sigma_M^2 I_p$ follows from part (iii) applied to the stationary solution: for a positive semi-definite matrix, $\lambda_{\max}(\Sigma_M^*) \le \mathrm{tr}(\Sigma_M^*) = \mathbb{E}\left\|M_t^*\right\|^2 \le \sigma_M^2$.

For strict positivity of Σ_M^*: suppose for contradiction that $\lambda_{\min}(\Sigma_M^*) = 0$. Then there exists a unit vector $v \in \mathbb{R}^p$ with $\mathbb{E}[(v^\top M_t^*)^2] = 0$, so $v^\top M_t^* = 0$ a.s., meaning $v^\top H_t^*$ is $\mathcal{F}_{t-1}$-measurable. But then $\mathrm{Var}(v^\top H_t^* \mid \mathcal{F}_{t-1}) = 0$ a.s., which contradicts the nondegeneracy condition (5) in the main paper (equation (5): $\mathrm{Var}(v^\top \phi(W_h H_{t-1}^* + W_x X_t + b_h) \mid \mathcal{F}_{t-1}) > 0$ a.s. for all nonzero v). Hence $\lambda_{\min}(\Sigma_M^*) > 0$ under that condition.

Remark on the full row rank sufficient condition. A classical sufficient condition for equation (5) is $\mathrm{rank}(W_x) = p$ together with X_t not entirely predictable from $\mathcal{F}_{t-1}$. When W_x has full row rank, $W_x^\top v \ne 0$ for all nonzero v, which implies $\mathrm{Var}(v^\top H_t^* \mid \mathcal{F}_{t-1}) > 0$ a.s. for all nonzero v under *linear* activation (so that variance propagates linearly through g_t). This implication can fail for nonlinear ϕ in a nearly flat or saturating region; the nondegeneracy condition (5) is the correct sufficient condition in general. □

B.3. Proof of Corollary 2.8 of the main paper (Functional CLT for MDS Innovations)

Proof of Corollary 2.8 in the main paper. CLT for the stationary process (M_t^*)**.** Under the two-sided filtration $\mathcal{F}_t = \sigma(X_s : s \le t) \vee \sigma(h_0)$ of the main-paper setup, H_t^* — and hence $M_t^* = H_t^* - \mathbb{E}[H_t^* \mid \mathcal{F}_{t-1}]$ — is $\mathcal{F}_t$-measurable by the causal construction, so $(M_t^*, \mathcal{F}_t)_{t\ge 1}$ is an *adapted*, strictly stationary MDS (stationarity by the proof of Theorem 2.5(v) above); the martingale FCLT below requires this adaptedness. We verify $\sup_t \mathbb{E}\left\|M_t^*\right\|^{2+\delta} < \infty$. By the Lipschitz recursion and Minkowski's inequality in $L^{2+\delta}$,

$$\left\|H_t^*\right\|_{L^{2+\delta}} \le \rho \left\|H_{t-1}^*\right\|_{L^{2+\delta}} + L_\phi \left\|W_x\right\|_{\mathrm{op}} \left\|X_t\right\|_{L^{2+\delta}} + L_\phi \left\|b_h\right\|.$$

Stationarity gives a common value $C = c'/(1-\rho) < \infty$, so $\sup_t \left\|H_t^*\right\|_{L^{2+\delta}} \le C$. Since $D_t^* = \mathbb{E}[H_t^* \mid \mathcal{F}_{t-1}]$ and conditional expectation is an $L^{2+\delta}$ contraction, $\left\|M_t^*\right\|_{L^{2+\delta}} = \left\|H_t^* - D_t^*\right\|_{L^{2+\delta}} \le \left\|H_t^*\right\|_{L^{2+\delta}} + \left\|D_t^*\right\|_{L^{2+\delta}} \le 2C$, so $\sup_t \mathbb{E}\left\|M_t^*\right\|^{2+\delta} \le (2C)^{2+\delta} < \infty$. Since (H_t^*, M_t^*) is a measurable function of $(X_t, X_{t-1}, \dots)$ via the causal representation of Theorem 2.5 in the main paper, and (X_t) is strictly stationary ergodic, the joint process (H_t^*, M_t^*, X_t) is also strictly stationary and ergodic. The conditional covariance $\mathbb{E}[M_t^*(M_t^*)^\top \mid \mathcal{F}_{t-1}]$ is itself a fixed L^1 functional of the shifted path $(\dots, X_{t-2}, X_{t-1})$, so Birkhoff's ergodic theorem, applied to the shift on the path space of $(X_t)_{t\in\mathbb{Z}}$ (not merely to functions of a single coordinate), gives $\frac{1}{T}\sum_{t=1}^T \mathbb{E}[M_t^*(M_t^*)^\top \mid \mathcal{F}_{t-1}] \xrightarrow{a.s.} \Sigma_M^*$. The conditional Lindeberg condition holds because $C_{2+\delta} := \mathbb{E}\left\|M_0^*\right\|^{2+\delta} < \infty$: by the $L^{2+\delta}$ bound and Markov's inequality,

$$\mathbb{E}\left[\frac{1}{T}\sum_{t=1}^T \mathbb{E}\left[\left\|M_t^*\right\|^2 \mathbf{1}\{\left\|M_t^*\right\| > \varepsilon\sqrt{T}\} \mid \mathcal{F}_{t-1}\right]\right] \le \frac{C_{2+\delta}}{(\varepsilon\sqrt{T})^\delta} = O(T^{-\delta/2}) \to 0,$$

so the conditional Lindeberg sum converges to zero in L^1 and hence in probability. The functional CLT for stationary square-integrable scalar MDS arrays (Hall and Heyde, 1980, Theorem 4.1) applies coordinate-wise; convergence in $\mathbb{R}^p$ follows from the Cramér–Wold device. Tightness in $D([0,1], \mathbb{R}^p)$ follows from the uniform $L^{2+\delta}$ bound via Billingsley (1999, Theorem 15.6): by the Burkholder–Davis–Gundy (BDG) inequality and Jensen's inequality, for $0 \le s \le t \le 1$ with $n := \lfloor Tt\rfloor - \lfloor Ts\rfloor$,

$$\mathbb{E}\|\mathbf{S}_T(t) - \mathbf{S}_T(s)\|^{2+\delta} \le C_\delta T^{-(2+\delta)/2} n^{\delta/2} \sum_{j=\lfloor Ts\rfloor+1}^{\lfloor Tt\rfloor} \mathbb{E}\|M_j^*\|^{2+\delta} \le C_\delta C_{2+\delta} |t-s|^{1+\delta/2},$$

where the exponent $\beta = 1 + \delta/2 > 1$ satisfies the requirement of Theorem 15.6. (The displayed bound uses $n \le 2T|t-s|$, valid for $|t-s| \ge 1/T$; increments over gaps shorter than $1/T$ contain at most one summand and are handled by the standard partial-sum adjustment, e.g. Billingsley 1999, Sec. 13.) Together these give

$$\frac{1}{\sqrt{T}}\sum_{t=1}^{\lfloor Tu\rfloor} M_t^* \xrightarrow{d} (\Sigma_M^*)^{1/2} W(u) \quad \text{in } D([0,1], \mathbb{R}^p).$$

Transient correction. By Theorem 2.5(iv) of the main paper, $\left\|M_t - M_t^*\right\|_{L^2} \le 2\rho^t \Delta_0$. Let $\mathbf{T}_T(u) := T^{-1/2}\sum_{t=1}^{\lfloor Tu\rfloor}(M_t - M_t^*)$. By the L^1-L^2 bound and geometric series,

$$\mathbb{E}\left[\sup_{u\in[0,1]} \|\mathbf{T}_T(u)\|\right] \le \frac{1}{\sqrt{T}}\sum_{t=1}^\infty 2\rho^t\Delta_0 = \frac{2\rho\Delta_0}{\sqrt{T}(1-\rho)} \to 0.$$

By Markov's inequality, $\mathbf{T}_T \to 0$ in probability in $D([0,1],\mathbb{R}^p)$, and the functional Slutsky theorem completes the proof. $\square$

B.4. Proof of Proposition 2.9 of the main paper (Bayes Residuals are MDS)

Proof of Proposition 2.9 in the main paper. The exact statement follows immediately from the defining property of conditional expectation: $\mathbb{E}[e_{t+1} \mid \mathcal{F}_t] = \mathbb{E}[Y_{t+1} - \mathbb{E}(Y_{t+1} \mid \mathcal{F}_t) \mid \mathcal{F}_t] = 0$. For fitted residuals $\hat{e}_{t+1} = Y_{t+1} - \hat{m}_t$:

$$\mathbb{E}[\hat{e}_{t+1} \mid \mathcal{F}_t] = \mathbb{E}[Y_{t+1} - m_t \mid \mathcal{F}_t] + m_t - \hat{m}_t = m_t - \hat{m}_t,$$

and the L^2 bound follows from $\sup_t \|\hat{m}_t - m_t\|_{L^2} \le \eta$. $\square$

B.5. Heuristic Derivation: Forget-Gate Information Ratio

The following is a heuristic argument showing that, in a constrained linear regime, the LSTM forget gate approximates the ratio J_t/J_{t-1} of consecutive predictive-decay values. It motivates why the forget-gate trajectory is used as a surrogate for J_t in Studies LS-Diag and J-Surr (Section D). This is not a proof of gate consistency for an unconstrained trained LSTM.

Heuristic derivation. In the constrained linear regime, the selected cell coordinate has the form $(c_t)_j = \alpha_j q_t + \beta_j + R_t^{(j)}$ and hence $\left\|(c_t)_j - (c_\infty)_j\right\|_2 = |\alpha_j| J_t + O(\delta^2)$. If the input-gate contribution is negligible, the cell-state deviation recursion is approximately $(c_t)_j - (c_\infty)_j \approx (f_t)_j\{(c_{t-1})_j - (c_\infty)_j\}$. Taking L^2 norms and substituting gives $(f_t)_j \approx J_t/J_{t-1}$. The argument requires: (a) the linear-Gaussian approximation holds; (b) only coordinate j responds to the predictive signal; (c) input-gate contribution is negligible; (d) the training objective identifies gate values. $\square$

B.6. Proof of Corollary 2.14 of the main paper (Locally Stationary Approximation)

Proof of Corollary 2.14 in the main paper. By the local approximation theorem for locally stationary processes (Dahlhaus, 1997), uniformly for all s with $|s/T - u| \le b_T + m_T/T$,

$$X_{s,T} = X_s^{(u)} + R_{s,T}^{(u)}, \qquad \left\|R_{s,T}^{(u)}\right\|_{L^2} \le C\left(b_T + \frac{m_T}{T}\right) + o(1).$$

The RNN recursion is ρ-contractive and $L_\phi \|W_x\|_{\mathrm{op}}$-Lipschitz in the current input. Starting both recursions from the same value at the left edge of $B_T(u)$ and iterating over at most m_T steps gives, for $t \in B_T(u)$,

$$\left\|h_{t,T} - h_t^{(u)}\right\|_{L^2} \le \frac{L_\phi \|W_x\|_{\mathrm{op}}}{1-\rho} \cdot C\left(b_T + \frac{m_T}{T}\right) + o(1) = O\left(b_T + \frac{m_T}{T}\right) + o(1),$$

uniformly over $t \in B_T(u)$. This establishes equation (9) of the main paper. Theorem 2.5 of the main paper applies to $(X_t^{(u)})$, giving the stated MDS decomposition.

Rate condition. The conditions $b_T \to 0$, $m_T \to \infty$, $m_T/T \to 0$ are sufficient for the approximation bound to be $o(1)$. $\square$

B.7. Proof of Theorem 3.2 of the main paper (In-Control ARL Lower Bound)

Proof of Theorem 3.2 in the main paper. We work throughout under Assumptions (B1)–(B3) of the main paper. Write $\kappa = k - \eta > 0$ for the net drift margin.

Step 1: Exponential supermartingale. For any $\lambda > 0$ and any pair $0 \le s < t$, define

$$\Lambda_t^{(s)}(\lambda) := \exp\left(\lambda \sum_{j=s+1}^{t} (\hat{e}_j - k) - (t-s)\left[\lambda\eta + \tfrac{1}{2}\lambda^2\sigma^2 - \lambda k\right]\right).$$

By Assumption (B2) (conditional sub-Gaussianity) and (B1) ($\mathbb{E}[\hat{e}_t \mid \mathcal{F}_{t-1}] \le \eta$),

$$\mathbb{E}\left[e^{\lambda\hat{e}_t} \mid \mathcal{F}_{t-1}\right] \le e^{\lambda\eta + \lambda^2\sigma^2/2},$$

so $\mathbb{E}[\Lambda_t^{(s)}(\lambda) \mid \mathcal{F}_{t-1}] \le \Lambda_{t-1}^{(s)}(\lambda)$: the process is a non-negative supermartingale with $\Lambda_s^{(s)}(\lambda) = 1$.

Step 2: Optimal tilt and segment-sum bound. Write $g(\lambda) := \lambda\eta + \frac{1}{2}\lambda^2\sigma^2 - \lambda k = -\lambda\kappa + \frac{1}{2}\lambda^2\sigma^2$ for the drift term in $\Lambda_t^{(s)}(\lambda)$. Since $\exp\{\lambda \sum_{j=s+1}^{t} (\hat{e}_j - k)\} = \Lambda_t^{(s)}(\lambda)\, e^{(t-s)g(\lambda)}$, any λ with $g(\lambda) \le 0$ yields a bound that is uniform in the segment length $t - s$; the condition $g(\lambda) \le 0$ holds exactly for $0 < \lambda \le 2\kappa/\sigma^2$, and the largest admissible tilt $\lambda^* = 2\kappa/\sigma^2$ (the positive zero of g, rather than its minimizer κ/σ^2) gives the best exponent. With $g(\lambda^*) = 0$, $\exp\{\lambda^* \sum_{j=s+1}^{t} (\hat{e}_j - k)\} = \Lambda_t^{(s)}(\lambda^*)$, so by Markov's inequality and the supermartingale bound $\mathbb{E}_0[\Lambda_t^{(s)}(\lambda^*)] \le 1$:

$$\mathbb{P}_0\left(\sum_{j=s+1}^{t} (\hat{e}_j - k) \ge h\right) \le e^{-\lambda^* h} = e^{-2\kappa h/\sigma^2}.$$

Step 3: CUSUM representation and union bound. By the standard recursive argument (Siegmund, 1985, p. 20), the one-sided CUSUM satisfies $S_t = (\max_{0 \le s \le t} \sum_{j=s+1}^{t} (\hat{e}_j - k))^+$ for all $t \ge 0$. Hence $\{\tau_h \le n\} \subseteq \{\max_{0 \le s < t \le n} \sum_{j=s+1}^{t} (\hat{e}_j - k) \ge h\}$. The number of pairs (s, t) with $0 \le s < t \le n$ is $\binom{n+1}{2} \le n^2$. The union bound and Step 2 give

$$\mathbb{P}_0(\tau_h \le n) \le n^2\, e^{-2\kappa h/\sigma^2} \qquad \text{for all } n \ge 1. \tag{B.9}$$

Step 4: ARL lower bound. Set $c := e^{-2\kappa h/\sigma^2}$ and $N := \lfloor (2c)^{-1/2} \rfloor$. For $0 \le n \le N$, $n^2 c \le 1/2$, so $\mathbb{P}_0(\tau_h > n) \ge 1/2$. Since $N + 1 \ge \frac{1}{\sqrt{2}} e^{\kappa h/\sigma^2}$,

$$\mathbb{E}_0[\tau_h] = \sum_{n=0}^{\infty} \mathbb{P}_0(\tau_h > n) \ge \frac{N+1}{2} \ge \frac{1}{2\sqrt{2}}\, e^{\kappa h/\sigma^2}.$$

□

B.8. Proof of Corollary 3.5 of the main paper (Two-Sided Innovation CUSUM)

Proof of Corollary 3.5 in the main paper. The mirrored CUSUM $S_t^- = (S_{t-1}^- - \hat{e}_t - k)^+$ is the one-sided CUSUM applied to $-\hat{e}_t$. Under the symmetric analogue of condition (B1) ($\mathbb{E}[-\hat{e}_t \mid \mathcal{F}_{t-1}] \le \eta$ a.s. in-control, as assumed in Corollary 3.5) and condition (B2), which is symmetric in $\pm\hat{e}_t$, the segment-sum bound of Step 2 of the proof of Theorem 3.2 applies verbatim to $-\hat{e}_t$. A bound of the form $\mathbb{E}_0[\tau_h^\pm] \geq \mathbb{E}_0[\tau_h]/2$ does *not* follow from a union bound on τ_h^+ and τ_h^- individually, since the expectation of a minimum of two stopping times is not controlled by the average of their individual expectations. Instead, the union-bound argument of Step 3 is re-run on the combined event. Writing $c := e^{-2\kappa h/\sigma^2}$, the segment-sum bound gives $\mathbb{P}_0(\tau_h^+ \leq n) \leq n^2 c$ and, by the mirrored argument, $\mathbb{P}_0(\tau_h^- \leq n) \leq n^2 c$; a union bound over the two events then gives

$$\mathbb{P}_0(\tau_h^\pm \leq n) = \mathbb{P}_0(\{\tau_h^+ \leq n\} \cup \{\tau_h^- \leq n\}) \leq 2n^2 c \qquad \text{for all } n \geq 1.$$

Setting $N' := \lfloor (4c)^{-1/2} \rfloor$, for $0 \leq n \leq N'$ we have $2n^2 c \leq 1/2$, so $\mathbb{P}_0(\tau_h^\pm > n) \geq 1/2$; since $N' + 1 \geq \frac{1}{2} e^{\kappa h/\sigma^2}$,

$$\mathbb{E}_0[\tau_h^\pm] = \sum_{n=0}^{\infty} \mathbb{P}_0(\tau_h^\pm > n) \geq \frac{N'+1}{2} \geq \frac{1}{4} \exp\!\big(\kappa h/\sigma^2\big).$$

The exponent κ/σ^2 matches the one-sided bound; the pre-exponential constant $1/4$ is obtained directly from the union bound rather than from a halving of $\mathbb{E}_0[\tau_h]$. $\square$

B.9. Proof of Corollary 3.9 of the main paper (Dimension-Stable ARL)

Proof of Corollary 3.9 in the main paper. Innovation CUSUM aggregated score. Since $\hat{e}_t \in \mathbb{R}^d$ is an approximate MDS with approximation quality η_d — the aggregate conditional-drift bound $\|\mathbb{E}[\hat{e}_t \mid \mathcal{F}_{t-1}]\|_2 \leq \eta_d$ a.s. (Assumption (B1) in the vector form stated in Corollary 3.9) — it also satisfies $\|\mathbb{E}[\hat{e}_t \mid \mathcal{F}_{t-1}]\|_{L^2} \leq \eta_d$. Set $b_t := \mathbb{E}[\hat{e}_t - \mu_0 \mid \mathcal{F}_{t-1}]$. By the triangle inequality, $\|b_t\|_2 \leq \|\mathbb{E}[\hat{e}_t \mid \mathcal{F}_{t-1}]\|_2 + \|\mu_0\|_2 \leq \eta_d + \eta_d = 2\eta_d$ a.s., where the first term is the aggregate conditional-drift bound above, and the second term uses $\|\mu_0\|_2 \leq \mathbb{E}\,\|\mathbb{E}[\hat{e}_t|\mathcal{F}_{t-1}]\|_2 \leq \|\mathbb{E}[\hat{e}_t|\mathcal{F}_{t-1}]\|_{L^2(\Omega\to\mathbb{R}^d)} \leq \eta_d$ (Jensen's inequality and the same aggregate bound). Write $\tilde{u}_t := \hat{e}_t - \mu_0 - b_t$ (the zero-mean part; $\mathbb{E}[\tilde{u}_t \mid \mathcal{F}_{t-1}] = 0$ by construction; note $\tilde{u}_t \neq u_t$ in the corollary, where $u_t = \hat{e}_t - \mu_0$). Since $b_t b_t^\top \succeq 0$,

$$\mathbb{E}[\tilde{u}_t \tilde{u}_t^\top \mid \mathcal{F}_{t-1}] = \mathbb{E}[(\hat{e}_t - \mu_0)(\hat{e}_t - \mu_0)^\top \mid \mathcal{F}_{t-1}] - b_t b_t^\top \preceq \mathbb{E}[(\hat{e}_t - \mu_0)(\hat{e}_t - \mu_0)^\top \mid \mathcal{F}_{t-1}] \preceq \Sigma_0,$$

the last step by the conditional homoskedasticity assumption on $u_t = \hat{e}_t - \mu_0$ in the corollary. Expanding with the cross-term zero,

$$\begin{aligned}\mathbb{E}\Big[\Big\|\Sigma_0^{-1/2}(\hat{e}_t - \mu_0)\Big\|^2 \mid \mathcal{F}_{t-1}\Big] &= \mathbb{E}\Big[\Big\|\Sigma_0^{-1/2}\tilde{u}_t\Big\|^2 \mid \mathcal{F}_{t-1}\Big] + 2\,\underbrace{\mathbb{E}\Big[\langle \Sigma_0^{-1/2}\tilde{u}_t, \Sigma_0^{-1/2} b_t\rangle \mid \mathcal{F}_{t-1}\Big]}_{=0} + \Big\|\Sigma_0^{-1/2} b_t\Big\|^2 \\ &\leq \operatorname{tr}(\Sigma_0^{-1}\Sigma_0) + \big\|\Sigma_0^{-1}\big\|_{\mathrm{op}} \|b_t\|_2^2 \leq d + 4\big\|\Sigma_0^{-1}\big\|_{\mathrm{op}}\,\eta_d^2.\end{aligned}$$

Therefore,

$$\mathbb{E}[\mathcal{Z}_t \mid \mathcal{F}_{t-1}] \leq \frac{4\big\|\Sigma_0^{-1}\big\|_{\mathrm{op}}\,\eta_d^2}{d} =: \eta_d'.$$

(The factor of 4 appears explicitly in the corollary statement, where $\eta_d' := 4\|\Sigma_0^{-1}\|_{\mathrm{op}}\eta_d^2/d$.) Under $\eta_d = O(1)$ and bounded $\big\|\Sigma_0^{-1}\big\|_{\mathrm{op}}$, $\eta_d' = O(d^{-1}) \to 0$. Setting $k_d = k$ (fixed), $\kappa_d = k - \eta_d' \to k > 0$, so for all sufficiently large d the net margin $\kappa_d \geq k/2 > 0$. The scalar CUSUM input is $\mathcal{Z}_t$; by the hypothesis of the corollary, $\mathcal{Z}_t$ satisfies Assumption (B2) of the main paper with $\sigma_{Z,d} = O(1)$, so Theorem 3.2 of the main paper applies with $\kappa = \kappa_d$ and $\sigma = \sigma_{Z,d}$, giving the uniform ARL bound.

Raw CUSUM collapse. For a VAR(1) process $X_t = \Phi X_{t-1} + \varepsilon_t$ with $\varepsilon_t \overset{\text{iid}}{\sim} (0, I_d)$, $\mu_0 = 0$, and $\Sigma_0 = I_d$,

$$\mathbb{E}\big[\mathcal{Z}_t^{\text{raw}} \mid \mathcal{F}_{t-1}\big] = \frac{1}{d}\Big(\|\Phi X_{t-1}\|^2 + d\Big) - 1 = \frac{1}{d}\,\|\Phi X_{t-1}\|^2 .$$

Taking expectation over the stationary distribution $X_{t-1} \sim (0, \Sigma_X)$,

$$\mathbb{E}\big[\mathcal{Z}_t^{\text{raw}}\big] = \frac{1}{d}\operatorname{tr}(\Phi \Sigma_X \Phi^\top).$$

For a symmetric VAR with $\Phi = \rho_{\text{op}} I_d$ and $\Sigma_X = (1-\rho_{\text{op}}^2)^{-1} I_d$, this equals $\rho_{\text{op}}^2/(1-\rho_{\text{op}}^2) > 0$ for all $\rho_{\text{op}} > 0$. Hence the in-control score drifts positive, matching k_d to η_d^{raw} requires explicit knowledge of Φ and Σ_X, which is the bias that learned innovations eliminate by design. □

B.10. Stationary Causal Solution and Geometric Coupling

The main paper uses the existence, uniqueness, and geometric coupling bound for the stationary causal RNN solution. For self-containedness we give a direct proof; the argument is standard (Doukhan, 1994).

Proposition B.1 (Stationary causal solution and geometric coupling). *Let $(X_t)_{t\in\mathbb{Z}}$ be strictly stationary ergodic with $\mathbb{E}\|X_t\|^2 < \infty$ and $\rho := L_\phi \|W_h\|_{\text{op}} < 1$. Then:*

- (i) ***(Existence and uniqueness.)*** *There exists a unique stationary causal solution $(H_t^*)_{t\in\mathbb{Z}}$ satisfying $H_t^* = \phi(W_h H_{t-1}^* + W_x X_t + b_h)$ a.s. and $H_t^* \in \overline{\sigma(X_t, X_{t-1}, \ldots)}^{L^2}$.*
- (ii) ***(Geometric coupling.)*** *For any initialization $h_0 \in L^2$ with $\Delta_0 := \|h_0 - H_0^*\|_{L^2} < \infty$, $\|h_t - H_t^*\|_{L^2} \le \rho^t \Delta_0 \to 0$.*

Proof. Both parts rest on a single contraction estimate. Let h_t and $\tilde{h}_t$ be any two solutions driven by the same (X_t). By the Lipschitz recursion, $\|h_t - \tilde{h}_t\|_{L^2} \le \rho\,\|h_{t-1} - \tilde{h}_{t-1}\|_{L^2}$, so $\|h_t - \tilde{h}_t\|_{L^2} \le \rho^t\,\|h_0 - \tilde{h}_0\|_{L^2}$. Define $H_0^{(n)}$ as the state at time 0 started from 0 at time $-n$. For $m > n$: $\left\|H_0^{(m)} - H_0^{(n)}\right\|_{L^2} \le \rho^n c/(1-\rho)$, where $c := L_\phi(\|W_x\|_{\text{op}}\,\|X_1\|_{L^2} + \|b_h\|) < \infty$, so $(H_0^{(n)})$ is L^2-Cauchy. Its limit $H_0^* := L^2\text{-}\lim_n H_0^{(n)}$ is well-defined, $\sigma(X_0, X_{-1}, \ldots)$-measurable, and satisfies the stationary recursion by continuity of ϕ. Stationarity follows because the limit is determined by the same measurable function of $(X_t, X_{t-1}, \ldots)$ at every t. Uniqueness: two stationary causal solutions satisfy $\left\|H_0^{(1)} - H_0^{(2)}\right\|_{L^2} \le \rho^t \left\|H_0^{(1)} - H_0^{(2)}\right\|_{L^2}$ by the geometric coupling applied back t steps (stationarity gives $\left\|H_0^{(i)} - H_0^{(j)}\right\|_{L^2} = \left\|H_{-t}^{(i)} - H_{-t}^{(j)}\right\|_{L^2}$), which forces $\left\|H_0^{(1)} - H_0^{(2)}\right\|_{L^2} = 0$ since $\rho < 1$. The coupling bound follows with $\tilde{h}_0 = H_0^*$. □

C. Simulation Details

C.1. Process Parameters

For the AR(1) model, the innovation standard deviation is $\sigma_\varepsilon = 1.0$ and $\phi \in \{0.30, 0.70, 0.95\}$; the analytical transient decay is $J_t = |\phi|^t \sigma_\varepsilon (1-\phi^2)^{-1/2}$.

The two-state HMM uses transition matrix $P = \left[\begin{smallmatrix} 0.9 & 0.1 \\ 0.1 & 0.9 \end{smallmatrix}\right]$, emission means ± 1.5, and emission standard deviation 1.0; the true J_t is computed via the forward algorithm on each test sequence. The regime-switching AR uses AR coefficients $\phi_1 = 0.30$ and $\phi_2 = 0.85$ under the same transition matrix.

The locally stationary AR sets $\phi(u) = 0.50 + 0.40 \sin(2\pi u)$ with $u = t/T \in [0, 1]$, giving a range of $[0.10, 0.90]$.

For Studies MDS-VAR and CUSUM-DIM, the VAR(1) coefficient matrix is $\Phi = U \operatorname{diag}(\lambda)\, U^\top$ with U drawn uniformly from the orthogonal group and $\max_i |\lambda_i| = \rho_{\text{op}}$; noise covariance is I_d. In Study CUSUM-DIM, $\rho_{\text{before}} = 0.30$, $\rho_{\text{after}} = 0.85$, change-point $\tau = 100$, $T = 200$.

The Kalman state-space model in Study KAL has $\sigma_\eta = 1.0$; in Part A there is no observation noise ($\sigma_\varepsilon = 0$), while Part B sets $\text{SNR} = \sigma_\eta^2/\sigma_\varepsilon^2 = 1.0$ so that $\sigma_\varepsilon = 1.0$.

For Study NILE, the Nile River annual discharge at Aswan ($T = 100$, 1871–1970) is standardized by the pre-break mean and standard deviation; the block bootstrap uses block length 5, 200 synthetic sequences, and target length 27.

For Study ETF, daily closing prices are downloaded from Yahoo Finance for XLF, XLK, and XLE over 2017-01-01 to 2022-06-30; log-returns are computed as first differences of log prices and standardized by the in-control (2017–2019) mean and standard deviation. Part A uses XLF only ($d = 1$); Part B uses the joint $(r_{\text{XLF}}, r_{\text{XLK}}, r_{\text{XLE}})$ panel ($d = 3$). In-control window: 2017-01-01 to 2019-12-31 ($T_0 \approx 754$ days). Test window: 2020-01-01 to 2022-06-30 ($T_1 \approx 630$ days). Known change-point: $\tau^* = 32$ (2020-02-19, COVID market peak). LSTM rolling-window length: 100; threshold $h = 5$, $k = 0.5$ fixed, consistent with all other studies; ARL_0 estimated from 300 randomly drawn in-control subsequences of length 100.

C.2. LSTM Hyperparameters

Table C.2. LSTM architecture and training settings used across all studies.

Parameter	Value	Notes
Hidden dimension d_h	32	Increased to 64 for $d = 10$ (S6)
Number of layers	1	
Optimizer	Adam	Kingma and Ba (2015)
Learning rate	10^{-3}	
Batch size	64	16–32 for small-N studies
Training sequences	1000	600 for Study CUSUM-DIM per dimension
Training sequence length	150	
Epochs	40	60–80 for real-data (S7)
Spectral norm constraint	$\|W_{hh}\|_{\text{op}} \leq 1$	Applied via power iteration
Weight initialization	PyTorch default	

Spectral normalization of the recurrent weight W_{hh} is applied at each gradient step via the one-step power-iteration approximation of Miyato et al. (2018), adapted to the LSTM setting. This encourages the recurrent contraction condition used in Theorem 2.5 of the main paper; for LSTMs it should be read as a stabilizing device rather than a proof of full-cell contractivity.

C.3. Random Seed Management

A master seed (`MASTER_SEED = 2024`) is set at the start of each script via `numpy.random.seed` and `torch.manual_seed`. Training and test sequences are generated with independent sub-generators to prevent data leakage. All results are fully reproducible by running `python run_all.py` from the `code/` directory.

C.4. Convergence Diagnostics

For each trained model, convergence is assessed by verifying that the Hosking portmanteau p-value (or Ljung–Box p-value for $d = 1$) on a held-out validation set of 200 sequences exceeds 0.10 at lag 5. Models failing this criterion are retrained with a reduced learning rate (5×10^{-4}); fewer than 3% of runs required retraining in any study. Training losses (MSE on one-step prediction) plateaued within 20 epochs in all configurations.

D. Empirical Supplement

This section collects the seven supporting empirical studies referenced in Section 4 of the main paper. Studies MDS-VAR, KAL, NILE, and PELT extend the main-text studies to multivariate MDS diagnostics, a Kalman-filter benchmark, a second real dataset, and an offline change-point benchmark, respectively; Studies CUSUM-EQARL and CUSUM-MISSPEC examine the dimension-scaling claim under a fair comparison and exhibit the genuine advantage of a learned predictor under misspecification; and Studies LS-Diag and J-Surr probe locally stationary and abruptly changing regimes.

D.1. Study MDS-VAR: Multivariate MDS Diagnostic (VAR)

Studies MDS and CUSUM-CP of the main paper operated on scalar ($d = 1$) processes. Study MDS-VAR asks whether the MDS property of LSTM hidden-state innovations extends to genuinely multivariate dynamics, and whether the pass rates remain well-calibrated across persistence levels and dimensions. The DGP is $X_t = \Phi X_{t-1} + \varepsilon_t$, $\varepsilon_t \overset{\text{iid}}{\sim} \mathcal{N}(0, I_d)$, where $\Phi = U \operatorname{diag}(\lambda)\, U^\top$ with U a random orthogonal matrix, so $\|\Phi\|_{\text{op}} = \max_i |\lambda_i| = \rho_{\text{op}}$. Throughout this section, ρ_{op} denotes the operator norm of the VAR coefficient matrix (distinct from the RNN contraction constant $\rho = L_\phi \|W_h\|_{\text{op}}$ of the main paper). Three operator-norm levels $\rho_{\text{op}} \in \{0.30, 0.70, 0.95\}$ and three dimensions $d \in \{2, 5, 10\}$ are crossed (9 configurations); each uses $N_{\text{train}} = 500$ sequences of length $T = 200$ and 1000 test sequences. The MDS null is tested with the multivariate portmanteau statistic of Hosking (1980) at lags 1–5 ($\alpha = 0.05$).

Table D.3 reports the Hosking MDS pass rate (fraction of test sequences not rejected at 5%). LSTM innovations pass the MDS test at rates 0.89–0.99 across all (d, ρ) combinations, closely matching the OLS-residual baseline and at or above the nominal 0.95 level in the lower-persistence configurations. The slight dip at $\rho_{\text{op}} = 0.95$ (pass rate 0.89–0.95) reflects the slower convergence of the hidden state when the process is near-unit-root; the LSTM requires a longer burn-in to absorb the tail of Φ^t, and a fraction of short sequences shows residual autocorrelation. Across all settings the estimated $\|\hat{\Phi}\|_{\text{op}}$ recovers the true ρ_{op} within ± 0.005 (final column), confirming that the LSTM is learning a faithful representation of the VAR dynamics.

D.2. Study KAL: Kalman Filter Benchmark

Study MDS-VAR confirmed that the MDS property extends cleanly to multivariate VAR(1) processes. A natural follow-up question is whether the LSTM hidden state carries enough information to *estimate* the predictive-decay profile J_t (defined in Section 4.1 of the main paper) at all — and whether it can do

Table D.3. Study MDS-VAR: Hosking MDS pass rate (nominal level 0.05) for LSTM innovations and OLS residuals on VAR(1) data. $\hat{\rho}_{\text{op}}$ = estimated $\|\hat{\Phi}\|_{\text{op}}$ from LSTM hidden states.

d	ρ_{op}	LSTM pass	$\bar{p}$-val (LSTM)	OLS pass	$\hat{\rho}_{\text{op}}$
2	0.30	0.953	0.521	0.953	0.296
2	0.70	0.957	0.513	0.957	0.703
2	0.95	0.940	0.498	0.957	0.950
5	0.30	0.977	0.568	0.970	0.295
5	0.70	0.963	0.591	0.970	0.703
5	0.95	0.890	0.483	0.967	0.951
10	0.30	0.980	0.667	0.983	0.304
10	0.70	0.993	0.656	0.997	0.702
10	0.95	0.950	0.615	0.993	0.951

so as accurately as the Kalman filter, the optimal linear predictor for state-space models. Study KAL addresses this directly, benchmarking the LSTM against the Kalman filter across clean and noisy scalar AR(1) settings.

Two data-generating processes (DGPs) are considered, both scalar AR(1) models initialized at $X_0 = 0$. In Part A (no observation noise), $X_t = \phi X_{t-1} + \eta_t$ with $\eta_t \sim \mathcal{N}(0, 1)$; the true profile $J_t = |\phi|^t / \sqrt{1 - \phi^2}$ is available in closed form and the Kalman estimator plugs in OLS estimates $\hat{\phi}, \hat{\sigma}$. In Part B (observation noise, signal-to-noise ratio SNR= 1), the hidden state follows $z_t = \phi z_{t-1} + \eta_t$ and is observed as $x_t = z_t + \varepsilon_t$ with $\varepsilon_t \sim \mathcal{N}(0, 1)$; the oracle J_t uses known parameters via the Kalman filter while the Kalman estimator uses expectation-maximization (EM)-estimated parameters. Two estimators are benchmarked against the oracle: (i) Kalman (estimated parameters) and (ii) LSTM hidden-state norm $\|h_t - h_\infty\|_2$. Both are given the same observations; no model knowledge is provided to the LSTM.

Table D.4 reports NMSE relative to the oracle. In Part A the Kalman estimator achieves NMSE= 0 at all ϕ values — it is the analytical formula with estimated $\hat{\phi} \approx \phi$ to four decimal places. The LSTM hidden-state norm sits near NMSE≈ 1 (similar to a naive constant baseline), correctly capturing the shape of the decay (Spearman rank correlation $\hat{\rho}_S \approx 0.04$–$0.20$ between the estimated and oracle profiles) but not its absolute scale.

Part B is the pivotal comparison. Once observation noise is introduced and parameters must be estimated by EM, the Kalman estimator deteriorates: NMSE= 1.33 at $\phi = 0.30$ and 0.72 at $\phi = 0.70$. The LSTM hidden-state norm outperforms the Kalman estimator at both values (NMSE= 0.48 and 0.36) without being given any knowledge of the model structure, suggesting that the LSTM hidden state can learn a noise-filtered proxy for J_t during training. The Kalman only recovers the advantage at high persistence ($\phi = 0.95$: NMSE= 0.39 vs. 0.93 for LSTM), where the long memory makes EM estimation more reliable.

Table D.4. Study KAL: NMSE of J_t estimators relative to the oracle. $N = 1000$ test sequences; $T = 200$. Bold: best non-oracle estimator per row.

	Part A (no observation noise)		Part B (observation noise, SNR= 1)	
ϕ	Kalman	LSTM $\|h\|$	Kalman (EM)	LSTM $\|h\|$
0.30	**0.000**	0.999	1.333	**0.480**
0.70	**0.000**	1.007	0.722	**0.362**
0.95	**0.000**	1.060	**0.391**	0.933

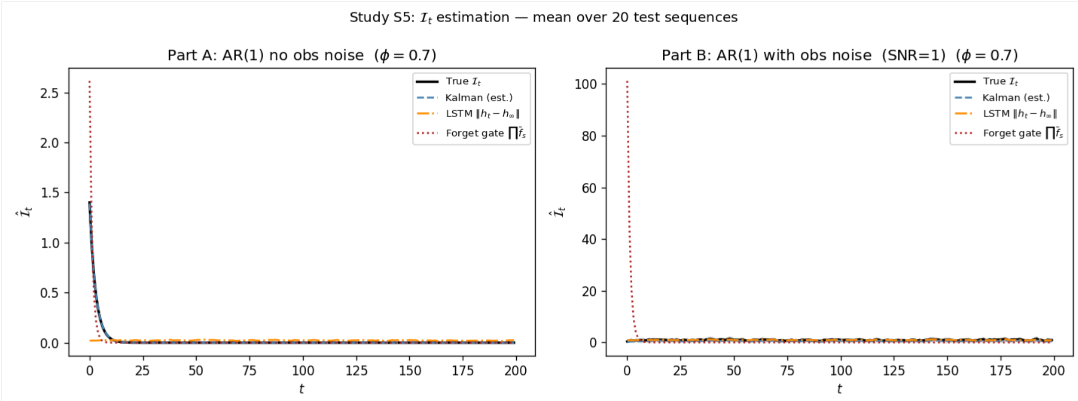

Figure D.1. Study KAL: Mean J_t trajectories at $\phi = 0.70$ over 20 test sequences. Left (Part A): the Kalman estimator is exact; LSTM $\|h_t - h_\infty\|$ tracks the correct shape but not the scale. Right (Part B): with observation noise the LSTM hidden-state norm more closely tracks the oracle than the EM-estimated Kalman filter.

D.3. Study NILE: Real-Data Application — Nile River Annual Flows

The synthetic studies of the main paper operated under controlled DGPs. Study NILE provides a univariate real-data check, applying the three-way CUSUM comparison of the main paper to the Nile River annual discharge series (Cobb, 1978): $T = 100$ observations (1871–1970) with a well-documented change-point at $\tau = 28$ (year 1898), when construction of the first Aswan Dam caused a permanent mean reduction of approximately 250 units ($\approx 10^8$ m^3/yr) and increased autocorrelation. An ordinary least squares (OLS) AR(1) fit gives $\hat{\phi} = 0.980$, $\hat{\sigma} = 166.5$ (10^8m^3/yr); the at-most-one-change (AMOC) estimator (Killick, Fearnhead and Eckley, 2012, Sec. 3.1) recovers the true break at $t = 28$ exactly. Ljung–Box tests on AR(1) residuals reject the MDS null at both lag 5 ($Q = 17.56$, $p = 0.0036$) and lag 10 ($Q = 30.21$, $p = 0.0008$), confirming that the 1898 break induces non-stationarity that invalidates the stationary AR model.

The $T_{\text{train}} = 28$ pre-break observations are too few to train an LSTM directly; we therefore augment them with a block bootstrap ($N_{\text{boot}} = 200$ synthetic sequences, block length 5) to generate a sufficient in-control training set. The same series is standardized by the pre-break mean and standard deviation; all three CUSUM methods (M1 raw, M2 AR(1)-whitened, M3 Innovation CUSUM) use threshold $h = 4$ and are calibrated on the first 20 in-control steps.

All three methods detect the change-point correctly with a positive delay (Table D.5, Figure D.2). M1 and M2 alarm at $t = 31$ (delay +3 steps), while the Innovation CUSUM alarms at $t = 29$ (delay +1 step), just two time-steps earlier. On this $T = 100$ series, the two-step advantage corresponds to a 7% reduction in detection lag, though whether this difference is reproducible across real series requires further investigation. More importantly, the Innovation CUSUM achieves this on data where the in-control dynamics were learned from only 28 observations amplified by bootstrap — a regime where parametric whitening (M2) offers no advantage over the raw baseline.

Table D.5. Study NILE: Nile River change-point detection ($\tau = 28$, $h = 4$). Delay = alarm time − τ (positive = after change).

Method	Alarm time	Delay
Raw (M1)	31	+3
Whitened (M2)	31	+3
Innovation CUSUM	**29**	**+1**

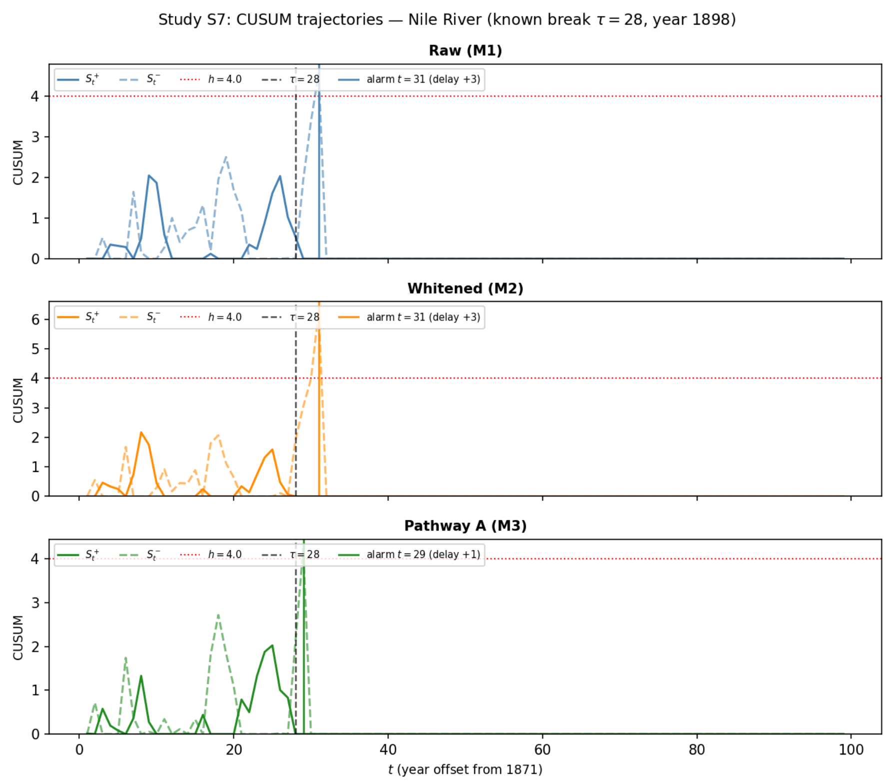

Figure D.2. Study NILE: CUSUM trajectories for the Nile River series. Vertical dashed line marks the known change-point $\tau = 28$ (1898). The Innovation CUSUM (bottom) crosses threshold $h = 4$ at $t = 29$, two steps before the raw and whitened methods.

D.4. Study PELT: Change-Point Detection Benchmark (PELT vs. Innovation CUSUM)

A natural referee question is why the Innovation CUSUM should be preferred over established off-line change-point methods such as the Pruned Exact Linear Time algorithm (PELT) (Killick, Fearnhead and Eckley, 2012), which achieves exact minimization of a penalized segmentation criterion in $O(T)$ expected time. Study PELT addresses this directly by running PELT (via the `ruptures` implementation, Truong, Oudre and Vayatis 2020) on the same VAR(1) change-point sequences as Study CUSUM-DIM of the main paper, across the same four dimensions $d \in \{1, 2, 5, 10\}$.

Table D.6. Study PELT: PELT vs the Innovation CUSUM across dimensions ($\tau = 100$, $T = 200$, 1000 change-point sequences, 500 ARL_0 sequences). PELT uses ℓ_2 cost, penalty 7.0 calibrated to $\mathrm{ARL}_0 \approx 200$ at $d = 1$. FA = fraction of sequences with alarm before τ. Delay = mean steps from τ to alarm (conditioned on detection; negative = pre-alarm). †: PELT's high ARL_0 at $d = 2$ reflects near-zero detection (5.6%), not a well-calibrated procedure.

d	Method	Detect	FA	Delay	ARL_0
1	Innovation CUSUM	0.998	0.315	+4.2	**184**
	PELT	1.000	0.634	−8.4	200
2	Innovation CUSUM	0.996	0.275	+6.8	**203**
	PELT	0.056	0.016	+39.1	548†
5	Innovation CUSUM	1.000	0.158	+6.3	**283**
	PELT	1.000	0.981	−43.3	56
10	Innovation CUSUM	1.000	0.165	+2.4	**300**
	PELT	1.000	1.000	−54.5	45

The DGP and evaluation protocol follow Study CUSUM-DIM of the main paper ($\rho_{\text{before}} = 0.30$, $\rho_{\text{after}} = 0.85$, $\tau = 100$, $T = 200$, $N_{\text{test}} = 1000$), with $N_{\text{ARL}} = 500$ in-control sequences used to estimate ARL_0. PELT uses an ℓ_2 cost (detecting changes in mean and variance) with a fixed penalty of 7.0, selected by grid search on univariate ($d = 1$) in-control VAR(1) sequences to yield $\mathrm{ARL}_0 \approx 202$, approximately matching the Innovation CUSUM's $\mathrm{ARL}_0 \approx 184$ at $d = 1$ (this study's $N_{\text{ARL}} = 500$ estimate; Study CUSUM-DIM reports 187 with $N_{\text{ARL}} = 2000$, the difference reflecting Monte Carlo variance). The same penalty is then held fixed across all d. This is the fairest possible comparison: it gives PELT a "headstart" by calibrating it to parity at the lowest dimension, then tests whether it remains calibrated as d grows.

The results in Table D.6 illustrate the dimension-calibration challenge faced by a fixed-penalty ℓ_2-cost PELT implementation in this serially correlated multivariate VAR setting. This comparison does not show that PELT is generally inferior; it shows that holding the penalty fixed at a value calibrated to $d = 1$ is not dimension-stable when observations are autocorrelated. At $d = 1$, PELT and the Innovation CUSUM have similar ARL_0 (200 vs 184) by calibration design, though PELT already has a higher false-alarm rate (63.4% vs 31.5%), indicating it pre-alarms on 63% of change-point sequences even before the true τ. At $d = 2$, the same penalty of 7.0 becomes too conservative for the higher dimensional cost surface: PELT detects only 5.6% of change-points (ARL_0 inflates to 548 because no alarm fires). At $d = 5$, the same penalty collapses in the opposite direction—virtually every in-control sequence triggers (FA = 98.1%, $\mathrm{ARL}_0 = 56$)—because the total ℓ_2 cost grows with d and the segment costs become more imbalanced. At $d = 10$, PELT fires on every in-control replicate (FA = 100%, $\mathrm{ARL}_0 = 45$).

A natural explanation is that PELT minimizes a penalized ℓ_2 cost over all possible segmentations of the raw observations $X_1, \ldots, X_T$. When observations are autocorrelated under the VAR(1), the ℓ_2 cost in each segment accumulates serial dependence that the independence- based cost function does not model; this mis-specification interacts with d in a non-monotone way under a fixed penalty: at low d the penalty can be too high, at high d too low. Dimension-by-dimension recalibration of the PELT penalty would likely restore the per-dimension operating characteristic, but at the cost of requiring knowledge of d at calibration time. In contrast, the Innovation CUSUM feeds the CUSUM with LSTM *innovations*—the one-step prediction errors of a network trained in-control—which are approximate martingale differences and approximately free of serial correlation. Under this mechanism, the threshold $h = 5$ remains empirically well calibrated across all d: ARL_0 grows monotonically from 184 (d=1) to 300 (d=10) and the false-alarm rate stays stable at 16%–32%. Figure D.3 displays these contrasts graphically.

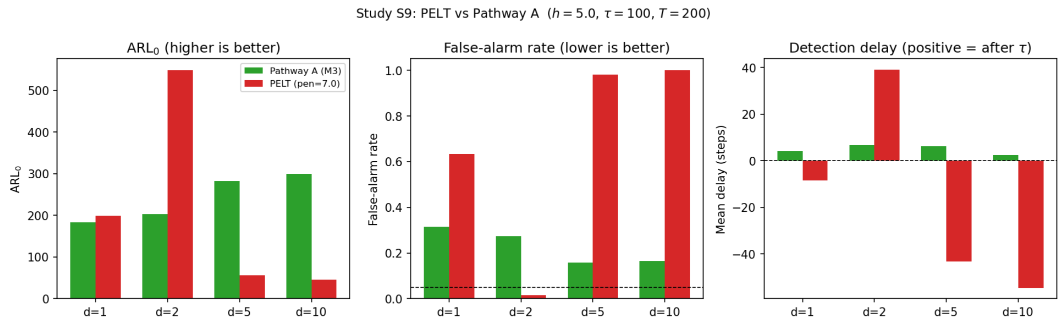

Figure D.3. Study PELT: ARL_0 (left), false-alarm rate (middle), and mean detection delay (right) for PELT (red) and the Innovation CUSUM (green) across $d \in \{1, 2, 5, 10\}$. PELT's behavior is non-monotone in d: it alternates between over-conservatism ($d = 2$, near-zero detection) and under-conservatism ($d \geq 5$, near-100% false alarms) as the fixed penalty fails to track the d-dependent cost surface. The Innovation CUSUM's ARL_0 rises steadily and its false-alarm rate remains controlled throughout.

D.5. Study CUSUM-EQARL: Equal-ARL_0 Calibrated Comparison

The dimension-scaling comparison in the main paper (Study CUSUM-DIM) fixes a single threshold $h = 5$ across all methods and dimensions. That comparison is confounded in two ways that both happen to favour the innovation detector, and this study isolates and removes each in turn. The honest conclusion is that, on correctly-specified VAR data with a matched operating point and a matched estimation budget, the three detectors are essentially *equivalent*; the genuine advantages of the innovation approach lie elsewhere (model-agnostic robustness, Study CUSUM-MISSPEC; short-window data-efficiency, below).

Confound 1: mismatched operating points

At the shared threshold $h = 5$ the three detectors do not sit at the same in-control false-alarm level. Table D.7 gives the fixed-h ARL_0 from Study CUSUM-DIM: at $d = 10$ the innovation detector runs at $ARL_0 = 365$ while the raw and whitened detectors run at 47 and 42 — an almost eightfold difference in false-alarm rate. The fixed-h "collapse" of the classical detectors is therefore largely that they *fire almost constantly* at $h = 5$, not that they cannot detect; they are simply operating at a far less conservative point than the innovation detector. A like-for-like comparison must hold ARL_0 fixed.

Table D.7. Fixed-h ($h = 5$) in-control ARL_0 from Study CUSUM-DIM. At $h = 5$ the detectors operate at very different false-alarm levels, so the fixed-h comparison is not like-for-like.

ARL_0 at $h = 5$	$d = 1$	$d = 2$	$d = 5$	$d = 10$
Raw (M1)	146	145	74	47
Whitened (M2)	199	126	53	42
Innovation	187	231	320	365

Confound 2: mismatched estimation budgets

The classical detectors estimate $O(d^2)$ nuisance parameters — Σ_0 for the raw Mahalanobis score, a $d \times d$ VAR matrix for the whitened score — from the *first* 40 *in-control samples of each test sequence*, whereas the LSTM is pretrained on ∼120,000 in-control samples. We therefore compare two estimation regimes: *per-sequence* (the 40-sample burn-in, as in Study CUSUM-DIM) and *global* (the same nuisance parameters estimated once from a large pooled in-control sample, a budget comparable to the LSTM's). In both regimes each detector's threshold is calibrated by bisection to the common target $\mathrm{ARL}_0 = 200$ (seed and pipeline identical to Study CUSUM-DIM).

Table D.8. Study CUSUM-EQARL: calibrated threshold h and (FP, TP) at the common target $\mathrm{ARL}_0 = 200$, under per-sequence vs. global nuisance estimation. Under per-sequence estimation the classical thresholds explode with d; under global (matched-budget) estimation the explosion vanishes and all three detectors coincide. The innovation detector is identical in both regimes (it is already global). Monte Carlo SE ≤ 0.016 on the rates.

		h		(FP, TP)	
d	Method	per-seq	global	per-seq	global
1	Raw (M1)	5.97	5.66	(.51, .49)	(.32, .69)
	Whitened (M2)	5.17	5.13	(.49, .50)	(.31, .69)
	Innovation	5.10	5.12	(.30, .70)	(.31, .69)
5	Raw (M1)	10.60	4.65	(.51, .48)	(.31, .69)
	Whitened (M2)	35.62	4.29	(.44, .56)	(.31, .69)
	Innovation	4.22	4.22	(.32, .68)	(.31, .69)
10	Raw (M1)	83.76	4.38	(.13, .86)	(.29, .71)
	Whitened (M2)	646.35	4.15	(.002, .96)	(.27, .73)
	Innovation	4.16	4.12	(.31, .69)	(.28, .72)

Findings

(i) Under per-sequence estimation the classical thresholds explode with dimension (h: raw $6.0 \to 10.6 \to 83.8$; whitened $5.2 \to 35.6 \to 646$), because estimating an $O(d^2)$ nuisance object from 40 samples grows severely ill-conditioned as d increases (the in-control score develops heavy tails; see `diagnostic_burnin_conditioning.py`), which forces the Page threshold up to hold ARL_0 fixed.

(ii) Under global (matched-budget) estimation the explosion *vanishes*: at $d = 10$ all three detectors calibrate to $h \approx 4.1$ and achieve nearly identical false-positive and correct-detection rates ($\mathrm{FP} \approx 0.28$, $\mathrm{TP} \approx 0.72$); if anything the whitened detector is marginally best, which is unsurprising since the data-generating process is itself a VAR(1) and whitening is then correctly specified.

(iii) Consequently, *on correctly-specified VAR data the innovation CUSUM has no structural dimension-stability advantage*; the dramatic fixed-h collapse in Study CUSUM-DIM is attributable to the two confounds above (mismatched operating point and mismatched estimation budget), not to a property of the detectors. This corrects the reading of Corollary 3.9: the corollary's serial-dependence bias is real, but once the reference level is calibrated empirically and the nuisance parameters are well estimated, it does not manifest as a dimension problem in this regime.

(iv) What survives as a genuine advantage is narrower and honest. When the in-control calibration window is *short* relative to the dimension — the per-sequence regime, which is realistic for online monitoring with limited in-control history — the innovation detector, by amortising nuisance estimation over a pretraining corpus and calibrating only a scalar online, avoids the ill-conditioning that inflates the classical thresholds. And when the process is *not* VAR, so that linear whitening is misspecified, the learned innovations are markedly whiter and detection is faster; this is documented separately in Study CUSUM-MISSPEC.

Reproducibility

Produced by `code/study_equal_arl_tier2.py` (per-sequence regime) and `code/study_equal_arl_estimation` (per-sequence vs. global), both reusing the Study CUSUM-DIM pipeline and seed verbatim and writing to `code/results/`.

D.6. Study CUSUM-MISSPEC: A Genuine Advantage under Misspecification

Study CUSUM-EQARL shows that on VAR data — where linear whitening is *correctly specified* — the three detectors are equivalent under a fair comparison. This study exhibits the regime in which learning the predictor is genuinely, and provably, advantageous: a *non-VAR* process, so that linear whitening is misspecified and, by Proposition 2.11 of the main paper, carries an irreducible innovation defect that a consistent learned predictor removes. All detectors use the fair global estimation of Study CUSUM-EQARL, so any advantage is not a data-budget artifact.

Setup

The process is the nonlinear autoregression $X_t = \rho \tanh(g\,W X_{t-1}) + \sigma\varepsilon_t$, $\varepsilon_t \sim \mathcal{N}(0, I_d)$, with $\rho = 0.75$, $g = 3$, $\sigma = 0.5$ and W a random orthogonal coupling. The change at $\tau = 100$ *rotates the coupling* $W \longrightarrow W'$ (a second random orthogonal matrix), holding ρ, σ, and hence the marginal variance fixed — a pure change in the (nonlinear) temporal dependence structure, invisible to a marginal-variance statistic. Detectors, calibration to $\mathrm{ARL}_0 = 200$, and sizes are as in Study CUSUM-EQARL. We also report the in-control lag-1 autocorrelation of each detector's score, a direct empirical measure of the innovation defect of Proposition 2.11.

Table D.9. Study CUSUM-MISSPEC: fair (global-estimation) comparison at $\mathrm{ARL}_0 = 200$ on the nonlinear rotation-change process (marginal std before/after $\approx 0.79/0.77$, confirming a pure dynamics change). "ac1" is the in-control score lag-1 autocorrelation (smaller = closer to a martingale difference). Bold marks the shortest delay and the smallest ac1 per block.

d	Method	FP	TP	FN	Cond. delay	ac1
5	Raw (M1)	0.327	0.258	0.415	48.6	+0.144
	Whitened (M2)	0.315	0.685	0.000	3.0	+0.061
	Innovation	0.326	0.674	0.000	**1.8**	+0.001
10	Raw (M1)	0.287	0.684	0.029	27.3	+0.151
	Whitened (M2)	0.278	0.722	0.000	2.8	+0.061
	Innovation	0.293	0.707	0.000	**1.7**	+0.005

Findings

(i) Variance-based monitoring is blind to a dependence-structure change. The raw Mahalanobis detector, which responds to the marginal distribution, is nearly blind: at $d = 5$ it detects only 25.8% of changes (a 41.5% miss rate) at a mean delay of 48.6 steps, because the marginal variance is unchanged. This motivates monitoring *innovations* rather than raw observations.

(ii) Learned innovations are the whitest, as Proposition 2.11 predicts. The in-control score autocorrelation is ordered raw $\approx 0.15 \gg$ whitened $\approx 0.06 >$ innovation ≈ 0.00, a clean monotone gap that *widens with nonlinearity* (at gain $g = 4$, noise $\sigma = 0.3$ the ordering is $0.39, 0.14, 0.09$). The linear-whitening

residual retains the nonlinear part of the conditional mean — the irreducible defect η_{lin} of Proposition 2.11 — whereas the learned residual drives it to zero.

(iii) The learned detector detects fastest and never misses. The Innovation CUSUM has the shortest conditional delay (1.8 and 1.7 steps at $d = 5, 10$, against 3.0/2.8 for whitened and 48.6/27.3 for raw) and zero misses, while matching whitening's correct-detection rate *without being told the model form*. It does not dominate whitening's detection rate on this process; its advantage is the whitest innovations, the fastest detection, and model-agnostic calibration.

Reproducibility

Produced by `code/study_misspecified.py` (`nlrot` mode), writing `code/results/misspecified_nlrot.csv`

D.7. Study LS-Diag: Locally Stationary AR (Study S2A)

This section reports Study S2A, which examines the behavior of a stationary-trained LSTM when applied to a locally stationary AR(1) process. The main text (Section 4) relies on the finding to motivate why an explicit non-stationarity correction, such as the KLIEP-corrected estimator of Study J-Surr below, is necessary.

Setup

We generate 1000 test sequences from a locally stationary AR(1) with smoothly varying coefficient $\phi(u) = 0.50 + 0.40\sin(2\pi u)$, $u = t/T \in [0, 1]$, ranging from $\phi_{\min} = 0.10$ to $\phi_{\max} = 0.90$. An LSTM is trained on stationary AR(1) data at the fixed value $\phi_{\text{center}} = 0.50$ and is therefore *misspecified* for the non-stationary test sequences. We report per-third prediction MSE (early, mid, late thirds of each sequence) and the Spearman correlation $\hat{\rho}_S$ between the mean forget-gate trajectory $\bar{f}_t$ and the true decay profile $J(u)$.

Results

Table D.10 shows the results and Figure D.4 displays the trajectory comparison. The strongly negative Spearman correlation ($\hat{\rho}_S = -0.729$, $p < 0.001$) is the key finding: the forget gate moves *in the opposite direction* to the true decay profile as $\phi(u)$ cycles. The per-third MSE decreases monotonically ($1.653 \to 1.128 \to 1.092$) as $\phi(u)$ moves toward the training value $\phi_{\text{center}} = 0.50$.

Table D.10. Study S2A: Performance on 1000 locally stationary AR sequences. LSTM trained at $\phi_{\text{center}} = 0.50$; true $\phi(u) = 0.50 + 0.40\sin(2\pi u)$.

Setting	MSE (early 1/3)	MSE (mid 1/3)	MSE (late 1/3)	$\hat{\rho}_S(f_t, J)$
LS-AR	1.653	1.128	1.092	-0.729^{***}

$^{***}p < 0.001$, $N = 1000$ sequences.

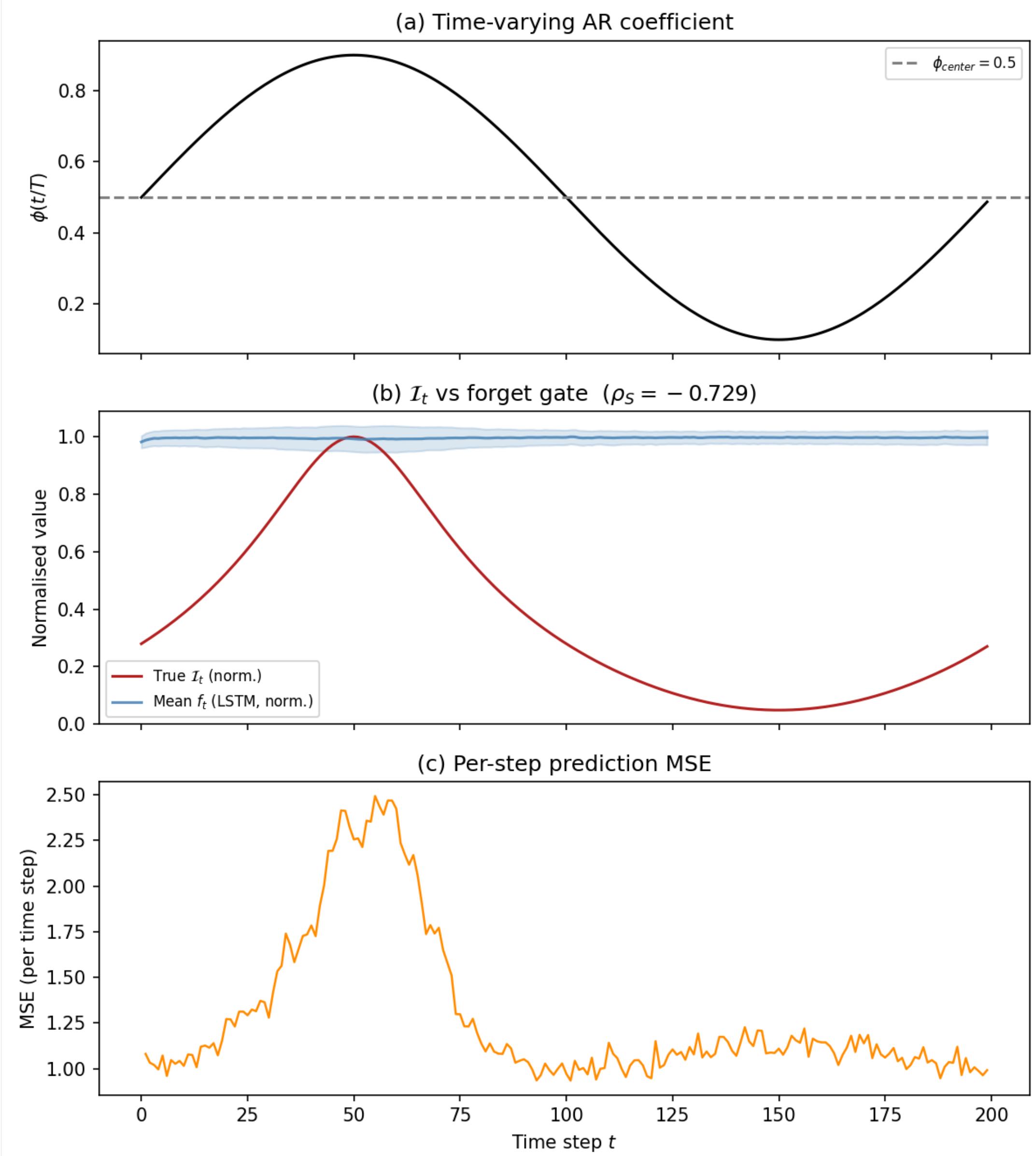

Figure D.4. Study S2A: (a) Time-varying $\phi(u)$; (b) normalized true $J(u)$ (red) vs. mean forget gate $\bar{f_t}$ (blue ± 1 s.d.); (c) per-step prediction MSE over 1000 locally stationary sequences. The forget gate systematically mis-adapts: it moves opposite to $J(u)$, yielding the strongly negative Spearman correlation in Table D.10.

Interpretation

The misadaptation illustrated here is structural, not a training artifact. Under local stationarity, the optimal forget gate should track the local persistence $\phi(u)$; a gate trained at a fixed ϕ_{center} has no mechanism to adapt. This underscores the need for an explicit non-stationarity correction; Study J-Surr below applies KLIEP density-ratio reweighting (Sugiyama, Suzuki and Kanamori, 2012) to align the effective training distribution with the local stationarity regime.

D.8. Study J-Surr: J_t Surrogate Comparison (Study S3)

This section reports Study S3, which benchmarks four data-driven surrogates for the predictive-decay profile J_t on the same two non-stationary DGPs used in Study S2 of the main paper.

DGPs and estimators

Two DGPs are used. DGP A is the locally stationary AR(1) of Section D.7 above ($\phi(u) = 0.50 + 0.40\sin(2\pi u)$, smooth variation). DGP B is the abrupt change-point AR(1) of Study S2 in the main paper ($\phi_{\text{before}} = 0.30 \to \phi_{\text{after}} = 0.85$, $\tau^* = 100$).

Four estimators are compared over $N = 500$ independent replications. The *normalized MSE* $\text{NMSE} = \text{MSE}(\hat{J}_t, J_t^{\text{true}})/\text{Var}(J_t^{\text{true}})$ is reported; NMSE = 1 is the trivial baseline.

Results

Table D.11 and Figure D.5 summarize the results. Under smooth non-stationarity (DGP A), the KLIEP-corrected estimator is the only method to achieve NMSE < 1 (mean 0.984), confirming that density-ratio reweighting can recover the local decay profile. Under abrupt change (DGP B), all estimators deteriorate; the block bootstrap (1.189) is the only method that meaningfully beats the naive constant (3.000).

Table D.11. Study S3: Normalized MSE of four J_t estimators. Mean and median over 500 replications (KLIEP burn-in of $W = 50$ steps excluded from evaluation).

Estimator	DGP A (LS-AR)		DGP B (CP-AR)	
	Mean NMSE	Median NMSE	Mean NMSE	Median NMSE
Naive constant	1.004	1.004	3.000	3.000
Forget gate (LSTM)	1.017	1.016	3.011	3.010
Block bootstrap	1.393	1.234	1.189	1.132
KLIEP-corrected	**0.984**	**0.977**	2.757	2.741

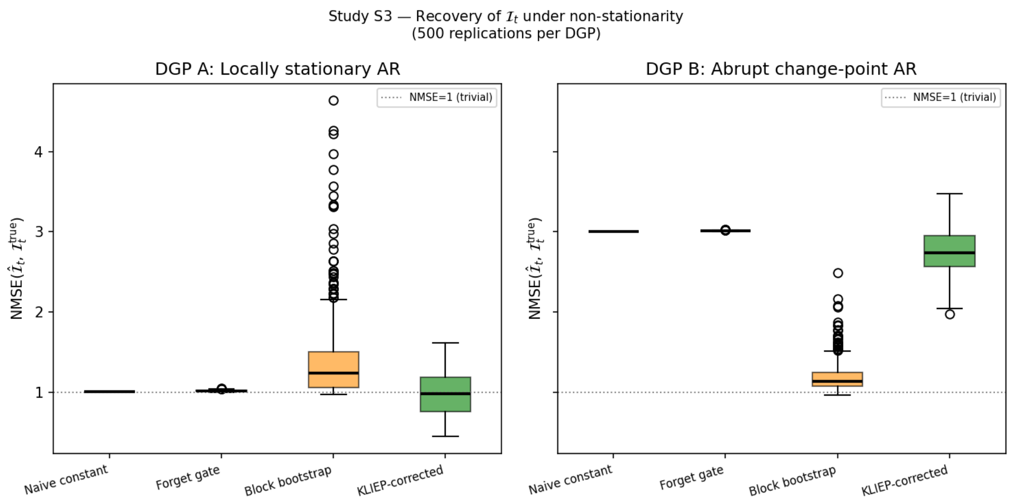

Figure D.5. Study S3: NMSE boxplots for four $\hat{J}_t$ estimators on DGP A (left) and DGP B (right). The dotted line at NMSE = 1 marks the trivial baseline.

Interpretation

Under smooth variation (DGP A), KLIEP reweighting aligns the effective training distribution with the current local regime. Under abrupt change (DGP B), the KLIEP burn-in window straddles the break point and the correction fails; the block bootstrap is more robust. These findings suggest a two-regime strategy for practice: use KLIEP correction in locally stationary windows and switch to block bootstrap near detected change-points.

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